

Continuous Optimization

Mathematical Models For Data Science

Pratyush Dayal

Indian Institute of Technology Gandhinagar

Introduction

Purpose

Basic numerical methods for training machine learning models.

Main branches of continuous optimization

- ① Unconstrained optimization
- ② Constrained optimization
- ③ **Special class:** Convex optimization

Assumptions

- ① Objective function is differentiable.
- ② Aim: minimize objective function—best value is minimum value (convention).

Limits, continuity and differentiability

Some questions to think about

- 1 Is the expression, $\frac{x-1}{x-1} = 1$ an identity? If no, then why $\lim_{x \rightarrow 1} \frac{x-1}{x-1} = 1$?
- 2 What is a continuous function? Recall the mathematical definition.
- 3 What is a differentiable function? How do you represent it mathematically?
- 4 Are all continuous functions differentiable? Are all differentiable functions continuous?
- 5 What is an objective function? What is a constraint?
- 6 Are maxima/minima for discontinuous functions defined? If so, how?
- 7 What is the curvature of a function? How is it found out?

Stationary points

Definition

- Stationary points are the real roots of the (first) derivative.
- Points where gradient is zero.

Example

To calculate the stationary points of a function: $\ell(x) = x^4 + 7x^3 + 5x^2 - 17x + 3$.

- 1 Determine the first derivative (gradient): $\frac{d\ell(x)}{dx} = 4x^3 + 21x^2 + 10x - 17$.
- 2 Calculate stationary points

$$4x^3 + 21x^2 + 10x - 17 = 0 \implies x \in [-4.5, -1.4, 0.7].$$

Extremum points

Definition

Stationary points that are either maxima or minima.

Question: Are all stationary points extremum points? Examples?

Finding minima

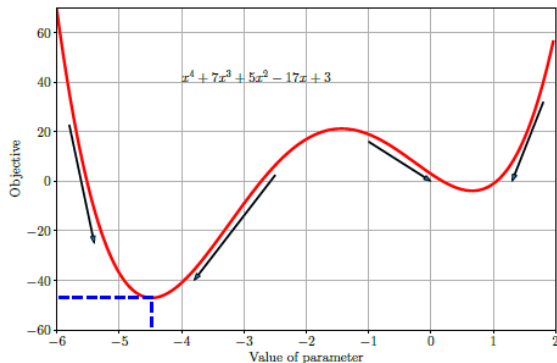
- 1 Calculate second derivative at all the stationary points.
- 2 Minima: second derivative is positive, Maxima: second derivative is negative.

Previous example

Second derivative $\frac{d^2\ell(x)}{dx^2} = 12x^2 + 42x + 10$. Thus, $x = -1.4$ is maxima $\left(\frac{d^2\ell(x)}{dx^2} < 0\right)$.

Global vs. local minima

Graphical example



- Arrows: Negative gradients
- Dashed line: Global minima

Optimization using gradient descent

Method

Solving for the minimum of a real-valued function $\min_{\mathbf{x}} f(\mathbf{x})$

- Assume f is differentiable, and its solution in closed form cannot be found analytically.
- Gradient descent is a first-order optimization algorithm and can be relatively slow close to minimum.
- The gradient points in the direction of the steepest ascent and that is orthogonal to the contour lines of the objective function.
- Contour lines: Lines connecting same values. Isobars, isotherms etc. are examples of contour lines.

Mathematical framework

Algorithm

- If for small step-size $\gamma \geq 0$, $f(\mathbf{x}_1) \leq f(\mathbf{x}_0)$ for

$$\mathbf{x}_1 = \mathbf{x}_0 - \gamma ((\nabla f)(\mathbf{x}_0))^T$$

- To find a local optimum $f(\mathbf{x}_*)$ of a function f , initial guess, $\mathbf{x}_0$ is chosen and then, iteration is performed according to

$$\mathbf{x}_{i+1} = \mathbf{x}_i - \gamma_i ((\nabla f)(\mathbf{x}_i))^T$$

- For suitable step-size γ_i , the sequence $f(\mathbf{x}_0) \geq f(\mathbf{x}_1) \geq \dots$ converges to a local minimum.

Numerical example I

Consider a quadratic equation in two dimensions

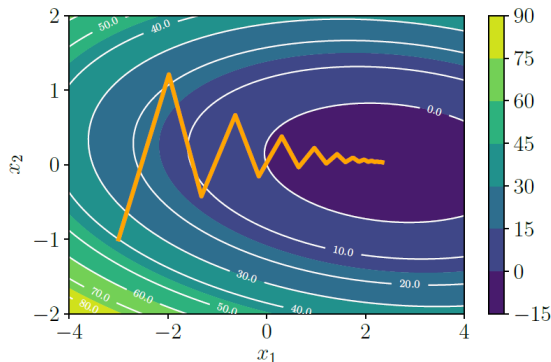
$$f\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \frac{1}{2} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}^T \begin{bmatrix} 2 & 1 \\ 1 & 20 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - \begin{bmatrix} 5 \\ 3 \end{bmatrix}^T \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

with gradient

$$\nabla f\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}^T \begin{bmatrix} 2 & 1 \\ 1 & 20 \end{bmatrix} - \begin{bmatrix} 5 \\ 3 \end{bmatrix}^T$$

Starting from $\mathbf{x}_0 = [-3, -1]^T$, we iterate to obtain a sequence of estimates that converge to the minimum value.

Graphical representation



The successive values of x are:

$$\mathbf{x}_0 = [-3, -1]^T,$$

$$\mathbf{x}_1 = [-1.98, 1.21]^T,$$

$$\mathbf{x}_2 = [-1.32, -0.42]^T \dots$$

Significance of step-size I

Step-size, γ , is important because:

When the function value

- **increases** after a gradient step, γ is too large. Undo the step and decrease γ .
- **decreases** the step could have been larger. Try increasing γ .

Significance of step-size II

Solving system of linear equations

Solution of $\mathbf{Ax} = \mathbf{b}$ is recast as solving $\mathbf{Ax} - \mathbf{b} = 0$. Objective is to find $\mathbf{x}_*$ that minimizes the squared error

$$\|\mathbf{Ax} - \mathbf{b}\|^2 = (\mathbf{Ax} - \mathbf{b})^T (\mathbf{Ax} - \mathbf{b})$$

represented as Euclidean norm (analogous to distance). The gradient with respect to $\mathbf{x}$ is

$$\nabla_x = 2(\mathbf{Ax} - \mathbf{b})^T \mathbf{A}$$

For numerical solution, use above expression in gradient descent. Otherwise, use analytical form.

Speed of convergence

For solution of linear systems of equations, gradient descent may converge slowly.

Condition number

Speed of of gradient descent is dependent on the condition number, $\kappa = \frac{\sigma(\mathbf{A})_{\max}}{\sigma(\mathbf{A})_{\min}}$ where, σ is singular value of $\mathbf{A}$.

Preconditioner

- Solve $\mathbf{P}^{-1}(\mathbf{A}\mathbf{x} - \mathbf{b}) = \mathbf{0}$, where $\mathbf{P}$ is the preconditioner.
- The goal is to design $\mathbf{P}^{-1}$ so that it has better κ than $\mathbf{A}$ and is easy to compute.

Gradient descent with momentum

Modifications

- 1 Introduces an additional term to remember what happened in the previous iteration. Dampens oscillations and smoothes out the gradient updates.
- 2 Remembers the update $\Delta \mathbf{x}_i$ at each iteration i and the next update is the linear combination of the current and previous gradients.
- 3 Averages out different noisy estimates of the gradient (moving average).

$$\mathbf{x}_{i+1} = \mathbf{x}_i - \gamma_i ((\nabla f)(\mathbf{x}_i))^T + \alpha \Delta \mathbf{x}_i$$

$$\Delta \mathbf{x}_i = \mathbf{x}_i - \mathbf{x}_{i-1} = \alpha \Delta \mathbf{x}_{i-1} - \gamma_{i-1} ((\nabla f)(\mathbf{x}_{i-1}))^T$$

Stochastic gradient descent

Modifications

- 1 SGD introduces a stochastic (noisy) approximation of the gradient for minimizing an objective function.
- 2 Stochastic approximation is written as a sum of differentiable functions.
- 3 In machine learning, given $n = 1, \dots, N$ data points, often the objective functions are the sum of the losses L_n incurred by each n , *i.e.*,

$$L(\boldsymbol{\theta}) = \sum_{n=1}^N L_n(\boldsymbol{\theta})$$

where, $\boldsymbol{\theta}$ is the parameters of interest that minimizes L .

Relationship with standard gradient descent I

Details

- Standard gradient descent is a “batch” optimization method, *i.e.*, optimization is performed using full training set by updating the vector of parameters as

$$\boldsymbol{\theta}_{i+1} = \boldsymbol{\theta}_i - \gamma_i (\nabla L(\boldsymbol{\theta}_i))^T = \boldsymbol{\theta}_i - \gamma_i \sum_{n=1}^N (\nabla L_n(\boldsymbol{\theta}_i))^T$$

for a suitable γ_i . May lead to **expensive evaluation** of gradients.

- In the RHS term, take a sum over a smaller set of L_n , *i.e.*, choose a subset of L_n (mini-batch). **Extreme case**: batch size is one.
- Justification**: RHS term is an empirical estimate of the expected value.

Relationship with standard gradient descent II

Implications in machine learning

- When the learning rate (rate of convergence) decreases at an appropriate rate, and subject to relatively mild assumptions, SGD converges almost surely to local minimum.
- Approximate gradient eases practical implementation constraints. Usage of CPU/GPU, memory size, computational time etc.

Convex Optimization

Advantage: Global optimality is guaranteed.

Convex optimization problem

- 1 $f(\cdot)$ is a convex objective function.
- 2 Constraints involving $g(\cdot)$ and $h(\cdot)$ are convex sets.

Convex set

- A set $\mathcal{C}$ is a convex set if for any $x, y \in \mathcal{C}$ and for any scalar θ with $0 \leq \theta \leq 1$, we have $\theta x + (1 - \theta)y \in \mathcal{C}$.
- Straight line connecting any two elements of the set lie inside the set.

Convex function

Definition

- Let function $f : \mathbb{R}^D \rightarrow \mathbb{R}$ be a function whose domain is a convex set.
- f is a convex function if for all x, y in its domain, and for any scalar θ with $0 \leq \theta \leq 1$, we have

$$f(\theta x + (1 - \theta)y) \leq \theta f(x) + (1 - \theta)f(y)$$

Relationships

- A concave function is the negative of a convex function.
- **Imagine** a convex function as a bowl-like object. Pour water to fill it up. Resulting set is **convex set** called **epigraph of the convex function**.

Differentiable convex functions

Relationship with gradient

If function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is differentiable then, it is convex if and only if

- 1 For any two points $\mathbf{x}, \mathbf{y}$

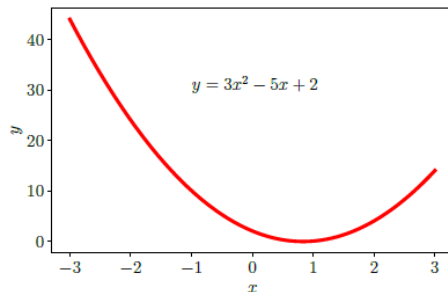
$$f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla_{\mathbf{x}} f(\mathbf{x})^T (\mathbf{y} - \mathbf{x})$$

also known as **Jensen's inequality**.

- 2 $\nabla_{\mathbf{x}}^2 f(\mathbf{x})$ is positive semidefinite.

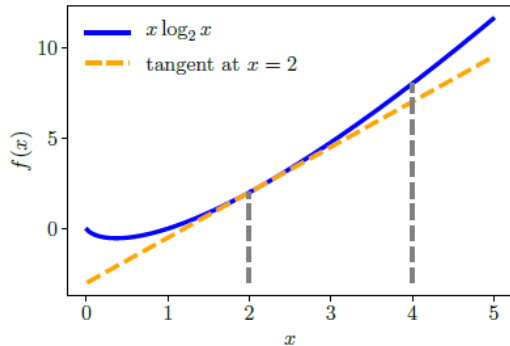
Note: f must be twice differentiable in this case.

Example



The negative entropy $f(x) = x \log_2 x$ is convex for $x > 0$.

- Illustrate definitions of convexity, using two points $x = 2$ and $x = 4$.
- To prove convexity of $f(x)$, all point x must be tested.



Calculating convexity of function

Using definition 1

- Consider $\theta = 0.5$ (midway between two points).
- Then, $\text{LHS} = f(0.5 \cdot 2 + 0.5 \cdot 4) = 3 \log_2 3 \approx 4.75$
- $\text{RHS} = 0.5(2 \log_2 2) + 0.5(4 \log_2 4) = 1 + 4 = 5$.
- Thus, $f(\theta x + (1 - \theta)y) \leq \theta f(x) + (1 - \theta)f(y)$ is satisfied.

Using definition 2

- $f(x)$ is differentiable, the derivative of $f(x)$ is $\nabla_x(x \log_2 x) = \log_2 x + \frac{1}{\ln 2}$.
- $\text{LHS} = f(4) = 8$ & $\text{RHS} = f(2) + \nabla f(2) \cdot (4 - 2) = 2 + (1 + \frac{1}{\ln 2}) \cdot 2 \approx 6.9$.
- Thus, Jensen's inequality is satisfied.

Constrained Optimization

Generalization

Jensen's inequality is a whole class of inequalities for taking non-negative weighted sums of convex functions.

Definition

A constrained optimization problem is called a convex optimization problem if

- 1 $\min_{\mathbf{x}} f(\mathbf{x})$ is to be found.
- 2 Subjected to $g_i(\mathbf{x}) \leq 0 \forall i = 1, \dots, m$ and $h_j(\mathbf{x}) = 0 \forall j = 1, \dots, n$.

Note: All functions $f(\mathbf{x})$ and $g_i(\mathbf{x})$ are convex, and all $h_j(\mathbf{x}) = 0$ are convex sets.

Linear Programming

Special case of constrained optimization

All the functions are linear, *i.e.*,

- 1 Find $\min_{x \in \mathbb{R}^d} c^T x$
- 2 Subjected to $Ax \leq b$

Here, $A \in \mathbb{R}^{m \times d}$ & $b \in \mathbb{R}^m$. This is known as a **linear program** with d linear program variables and m linear constraints. Optimization problem $\Rightarrow$ **Primal problem**

Lagrange and Lagrange multiplier

$$\mathcal{L}(x, \lambda) = c^T x + \lambda^T (Ax - b) = (c + A^T \lambda)^T x - \lambda^T b$$

where, λ is the vector of non-negative Lagrange multipliers.

Primal vs. Dual Problem

Dual Lagrangian

- Derivative of $\mathcal{L}(\mathbf{x}, \lambda)$ with respect to $\mathbf{x}$ and setting it to zero gives: $\mathbf{c} + \mathbf{A}^T \boldsymbol{\lambda} = \mathbf{0}$.
- The dual Lagrangian is: $\mathcal{D}(\boldsymbol{\lambda}) = -\boldsymbol{\lambda}^T \mathbf{b}$.

Dual optimization problem

- 1 Find $\max_{\boldsymbol{\lambda} \in \mathbb{R}^m} -\mathbf{b}^T \boldsymbol{\lambda}$
- 2 Subjected to $\mathbf{c} + \mathbf{A}^T \boldsymbol{\lambda} = \mathbf{0}$
- 3 With $\boldsymbol{\lambda} \geq \mathbf{0}$

This is also a **linear program**, but with m variables.

Linear Programming Demo

Example

① Find $\min_{x \in \mathbb{R}^2} - \begin{bmatrix} 5 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ (Objective Function)

② Subjected to $\begin{bmatrix} 2 & 2 \\ 2 & -4 \\ -2 & 1 \\ 0 & -1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \leq \begin{bmatrix} 33 \\ 8 \\ 5 \\ -1 \\ 8 \end{bmatrix}$ (Constraints)

Graphical Solution

Key observations

- ① The objective function (contour) is linear.
- ② Constraints are legends.
- ③ Optimal value is $\star$.

Points to ponder

- ① What does the shaded area represent?
- ② Why is $\star$ the minimum value?
- ③ Can the minima lie in the shaded region?

Projection on x_1, x_2 plane

